Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/07/2026
PI Name:	Mehta, Ranjana Ph.D.		
Project Title:	Characterizing and Mitigating the Interactive Impacts of Fatigue- and Altered Gravity-Related Stressors on Sensorimotor, Behavioral, and Operational Outcomes		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks (3) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:		**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/01/2024	**End Date:**	12/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per C. Ribeiro/JSC (Ed., 7/24/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wood, Scott Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC24K0485		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Crews during lunar orbital and surface missions will be exposed to multiple spaceflight stressors simultaneously (such as fatigue and altered gravity), which can adversely impact their sensorimotor and behavioral functions and subsequently result in performance decrements on manual/vehicle control or extravehicular activities. Despite the growing evidence reported in terrestrial studies on the impacts (additive, synergistic, or antagonistic) of vestibular challenges and altered sleep/cognition on sensorimotor, behavioral, and performance outcomes, very few ground analog studies of human spaceflight studies have targeted multiple NASA Human Research Roadmap risks to characterize these interactions. This knowledge gap has limited the validation of countermeasures that can synergistically mitigate multiple risks. Additionally, following long-duration spaceflight, crewmembers often report an increase in fogginess. Yet the existing Spaceflight Standard Measures battery does not include outcome measures that can interpret the relative contributions of multiple hazards on sensorimotor declines. This proposal aims to characterize the interactive impacts of significant spaceflight hazards (i.e., cognitive overload, sleep deprivation, and altered gravity) on sensorimotor functions and associated adaptive or disruptive neural responses that explain these interactive effects. We will employ a within-subjects experimental design to evaluate the interactive effects of altered gravity and fatigue on sensorimotor standard measures. Participants will attend two separate and counterbalanced fatigue sessions: a 2-h cognitive fatigue session (Aim 1), and a total sleep deprivation (~28-h extended wakefulness) session (Aim 2). At various time points in each fatigue session, participants will undergo a series of sensorimotor tests under two altered gravity analog conditions: absence, and presence, of galvanic vestibular stimulation (GVS). The sensorimotor tests include (a) tandem walk, the obstacle walk, and the finger-nose motor-coordination task (based on NASA's Standard Measures for sensorimotor assessment and prior flight studies); (b) motor coordination and standardized balance tasks; and (c), two additional tasks (a dual-task tandem walk test and a hidden obstacle walk test, designed as transition walks between the sensorimotor tasks that participants will unknowingly perform). Across both aims, we will assess changes in performance, brain activation, and connectivity patterns, and workload perceptions using the NASA Task Load Index (TLX) questionnaire during the sensorimotor tasks. Study findings will facilitate (1) characterizing the interactions of fatigue and altered gravity on diverse types of sensorimotor functions (functional mobility vs. manual control); and (2) establishing new outcomes measures (i.e., variants of existing sensorimotor tests, workload perceptions) that may be included in the sensorimotor Spaceflight Standard Measures to elucidate these interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We will characterize the potential interactive impacts of multiple spaceflight hazards (fatigue, altered gravity), on sensorimotor, and operational outcomes and associated adaptive or disruptive neural responses that explain these interactive effects. We will establish additional outcome measures (e.g., workload perceptions; dual-task/DT tandem walk; hidden obstacle walk) that can assess the relative contributions of multiple stressors on sensorimotor impairments in astronauts.
Task Progress:	Data analysis has been completed for both aims. For cognitive fatigue, results showed that fatigue significantly impaired performance on several sensorimotor tasks, with the most pronounced effects observed in the tandem walking tasks that required the greatest cognitive engagement. Specifically, the eyes-open and dual-task tandem walks showed significant declines in performance after the fatigue-induction period. GVS independently reduced performance across nearly all tandem walking and postural balance tasks. The postural balance tasks performed on the foam surface, which require greater reliance on proprioceptive feedback, showed significant effects of both fatigue and GVS. In contrast, the hidden obstacle walk and eye-hand coordination tasks were less sensitive to fatigue, suggesting that not all sensorimotor functions are equally vulnerable to fatigue. Cortical activation data revealed widespread reductions in brain activity across premotor and prefrontal regions following cognitive fatigue, particularly during tandem walking. Importantly, a significant interaction between fatigue and GVS was identified in the supplementary motor area during the hidden obstacle walk, indicating that the combined stressors engaged distinct neural mechanisms not observed under either condition alone. Subjective workload ratings increased significantly under both fatigue and GVS conditions, with combined conditions producing the highest workload perceptions across most tasks. In sleep-deprived subjects, a preliminary analysis across the four time points indicates progressive declines in sensorimotor performance with increasing hours of wakefulness. Tandem walking performance and postural balance measures showed time-on-task effects consistent with the accumulation of sleep pressure. GVS continued to independently degrade sensorimotor performance across multiple task domains, and early results suggest that sleep deprivation may amplify the degree of vestibular disruption experienced during balance-demanding activities. Cortical activation patterns during sleep deprivation revealed a distinct profile from those observed under cognitive fatigue, with some regions showing compensatory increases in activation at intermediate time points before declining at later stages of sleep deprivation.
Bibliography Type:	Description: (Last Updated: 05/27/2026)
Abstracts for Journals and Proceedings	Nichols KM, Jayasuriya J, Hayes J, Fairchild B, Mehta RK. "The impacts of altered gravity and mental fatigue on sensorimotor assessments." 2024 American Society of Biomechanics (ASB) Annual Meeting, Madison, Wisconsin, August 5-7, 2024. Abstracts. 2024 American Society of Biomechanics (ASB) Annual Meeting, Madison, Wisconsin, August 5-7, 2024. , Aug-2024
Abstracts for Journals and Proceedings	Jayasuriya J, Easton A, Mehta RK. "Predicting gait instabilities under vestibular perturbations to inform assistive technologies for planetary mission." 2026 Human Factors and Ergonomics Society Annual Meeting, Reno, Nevada, October 19-23, 2026. Abstracts. 2026 Human Factors and Ergonomics Society Annual Meeting, Reno, Nevada, October 19-23, 2026. , Oct-2026
Abstracts for Journals and Proceedings	Jayasuriya JR, Mehta RK. "Neural dynamics associated with sensorimotor performance in an altered-gravity analog." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026

Abstracts for Journals and Proceedings	Easton A, Jayasuriya J, Mehta R. "Interactive effects of vestibular impairment and mental fatigue on fine motor coordination in spaceflight analogs." 2025 Human Factors and Ergonomics Society Annual Meeting, Chicago, Illinois, October 13-17, 2025. Abstracts. 2025 Human Factors and Ergonomics Society Annual Meeting, Chicago, Illinois, October 13-17, 2025. , Oct-2025
Articles in Peer-reviewed Journals	Shanghavi AA, Jayasuriya J, Cai Y, Mehta RK. "Pulse rate variability from wearable wristwatches as a surrogate for heart rate variability? We think not yet." Proc Hum Factors Ergon Soc Annu Meet. 2025 Jul 9;10711813251357929. https://doi.org/10.1177/10711813251357929 , Jul-2025